Project Proposal Form
Institute of Information Technology (IIT)
University of Dhaka

Student's Name: [Your Name] **Student's Roll:** [Your Roll] **Phone:** [Your Phone]

Project Title: DENSEWALK-NAV: Real-Time Navigation Assistant for Visually Impaired Persons in Dense Urban Environments

Project Description:
DENSEWALK-NAV is a wearable, real-time navigation assistance system designed for visually impaired individuals navigating dense, unstructured street environments common in the Global South. A chest- or glasses-mounted camera feeds live video into a multi-stage perception pipeline that produces spoken navigation instructions through an earphone in under 300 milliseconds.

The system integrates three concurrent perception models — Apple Depth Pro for metric depth estimation, RAFT-Small for optical flow, and YOLOv8n with ByteTrack for pedestrian detection and tracking. Derived signals including time-to-collision, agent motion class, gap corridors, and zone occupancy feed a deterministic rule-based instruction engine that generates short spoken warnings without any large language model in the critical path. A priority interrupt queue ensures imminent threats override scheduled speech immediately.

Motivation:
Existing assistive navigation tools are designed for structured environments with clear sidewalks and predictable pedestrian flow. Dhaka and similar dense urban settings present a fundamentally different challenge: unmarked roads, erratic traffic, high pedestrian density, and overhead obstacles. Standard GPS-based solutions offer no real-time obstacle awareness. Camera-based assistants relying on large vision-language models (VLMs) incur 1.5–4 second inference delays, rendering them unsafe for active navigation. DENSEWALK-NAV addresses this gap by replacing VLM inference with a fast deterministic rule engine, making sub-300ms guidance feasible on commodity edge hardware.

Project Scope & Features:
- Real-time camera capture loop at 30fps with keyframe extraction every 0.1s.
- Parallel perception pipeline: metric depth, optical flow, and multi-object tracking.
- Derived safety signals: time-to-collision (TTC), agent motion class, gap corridors, zone occupancy, ego speed.
- Deterministic rule-based NLG engine producing spoken instructions in under 5ms.
- Priority interrupt queue: critical threats (TTC < 2s) preempt any ongoing speech.
- TTS output via edge-tts targeting < 300ms end-to-end latency.
- Offline Qwen2-VL-7B harness for post-session edge-case audit and threshold tuning.
- Designed for deployment on Android phone, Raspberry Pi, or NVIDIA Jetson Orin.

Languages & Technologies:
- **Language:** Python
- **Perception Models:** Apple Depth Pro, RAFT-Small (torchvision), YOLOv8n + ByteTrack (ultralytics)

- **Instruction LLM (offline only):** Qwen2-VL-7B (4-bit quantized, HuggingFace)
- **TTS:** edge-tts / pyttsx3
- **Deep Learning Framework:** PyTorch, HuggingFace Transformers
- **Camera Interface:** OpenCV VideoCapture
- **Version Control:** Git, GitHub
- **IDE:** Visual Studio Code, Jupyter Notebook
- **Target Hardware:** Laptop (dev) → Android / Jetson Orin (prod)

Timeline:

Sl	Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Perception pipeline port (Depth Pro + RAFT + YOLO, real-time single-frame)																		
2	Rule engine & template NLG																		
3	Priority queue & interrupt logic																		
4	TTS integration & latency optimization																		
5	Edge device deployment & field testing																		
6	Offline VLM audit harness																		
7	Final report writing																		

Supervisor's Name: _____

Signature of the supervisor: _____

Date: _____

Proposal Presentation Feedback:

Midterm Presentation Feedback: